LG AKB76043105 Magic Remote — Button Function Specification

Detailed Button-Level Functional Requirements for LG 2023 Series TV Remote Control

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3. Document Revision History

Version	Date	Author	Description
V1.0	2025-09-14	SQE Team	Initial release for LG 2023 Series validation

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4. Product Overview

4.1 Product Identity

The LG AKB76043105 is the designated Magic Remote controller for the LG 2023 television lineup. Finished in a matte black texture with anti-fingerprint coating, this device serves as the primary human-machine interface for WebOS 23. It is engineered to support the full spectrum of smart TV functionalities, ranging from traditional broadcast tuning to complex gesture-based navigation and voice-driven search. This specification document defines the strict functional requirements for every physical and logical input mechanism on the device.

4.2 Key Features Summary

The AKB76043105 integrates multiple communication and sensing technologies to ensure a seamless user experience:

- **Hybrid Connectivity:** Simultaneous support for Bluetooth Low Energy (BLE) 5.0 for smart functions and Infrared (IR) 38kHz for legacy power/volume control of external audio devices.
- **Motion Sensing:** Integrated 3-axis MEMS gyroscope and 3-axis accelerometer for Point & Click cursor control and gesture recognition.
- **Voice Interface:** Dual MEMS microphone array with beamforming and acoustic echo cancellation (AEC) for hands-free and push-to-talk voice commands.
- **Haptic & Visual Feedback:** Tactile click mechanism on the scroll wheel and dynamic LED ring illumination for mode indication and battery status.
- **Universal Control:** HDMI-CEC (SimpLink) and IR learning capabilities to control set-top boxes, soundbars, and receivers without switching remotes.
- **Accessibility First:** Dedicated physical keys for Audio Description, Screen Reader, and Magnifier, compliant with WCAG 2.1 remote control guidelines.

4.3 Hardware Specifications

Parameter	Specification	Tolerance / Notes
Dimensions	41 mm (W) × 190 mm (H) × 31 mm (D)	±1.0 mm
Weight	96 g (including batteries)	±2 g
Power Supply	2 × AAA Alkaline Batteries	LR03 / AM-4
Bluetooth	BLE 5.0	Class 2
IR Carrier Frequency	38 kHz	±1 kHz
Motion Sensors	3-Axis Gyro + 3-Axis Accel	6-DOF MEMS
Microphone	Dual MEMS Array	SNR > 61 dB
Operating Temp	0°C to +40°C	Non-condensing
Storage Temp	−20°C to +60°C	Non-condensing
BT Effective Range	10 m	Line of sight
IR Effective Range	8 m	Line of sight
Button Life	500,000 cycles	Mechanical switch
Scroll Wheel Life	1,000,000 cycles	Optical encoder

4.4 Supported TV Models

This remote is validated for the following LG 2023 model families:

- **OLED:** C3, G3, B3, A3
- **QNED:** QNED90, QNED80, QNED75
- **UHD:** UR90, UR80, UR75, UR73
- **NanoCell:** NANO76, NANO75

Note: While IR functionality may operate on older models, full Magic Remote features (Pointer, Voice, Scroll) are restricted to the 2023 WebOS 23 platform.

4.5 Software/Firmware Compatibility

- **TV OS:** WebOS 23 (Build 03.30.xx and above)
- **Remote Firmware:** Magic Remote FW v2.0.0 or later
- **Pairing Protocol:** LG Magic Remote Pairing Service v4.1

4.6 Package Contents

1. LG AKB76043105 Magic Remote Unit × 1
2. AAA Alkaline Batteries (LR03) × 2
3. Quick Start Guide (Multi-language) × 1
4. Warranty & Safety Card × 1

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5. Remote Control Physical Layout Description

This section defines the spatial arrangement of all user interface elements. Test engineers must verify that physical button mapping corresponds to the logical zones described below.

5.1 Top Section: Power & Volume Zone

Located at the uppermost edge of the remote, this zone contains controls intended for quick access without visual confirmation.

- **Power Button:** Top-center, slightly recessed. Red tactile cap. Distinctive high-travel switch feel.
- **Volume Rocker:** Right-side vertical rocker, positioned 15mm below the Power button. Upper segment is VOL+, lower is VOL-.
- **Channel Rocker:** Left-side vertical rocker, mirrored position to Volume. Upper segment is CH+, lower is CH-.
- **Mute Button:** Centered between Volume and Channel rockers, 5mm below the Power button. Concave surface for tactile differentiation.

5.2 Upper Middle Section: System Navigation

This zone handles global system states and application layering.

- **Input Selector:** Top-left of this zone. Icon: Box with arrow.
- **Home Button:** Top-center of this zone. Icon: House. Slightly larger surface area than adjacent keys.
- **Settings Button:** Top-right of this zone. Icon: Gear.
- **Back Button:** Left side, below Input. Icon: Curved arrow left.
- **Exit Button:** Right side, below Settings. Icon: ‘X’ or Door.

- **Info Button:** Center, below Home. Icon: ‘i’ in circle.

5.3 Center Section: Pointer & Selection Core

The ergonomic center of gravity and primary interaction zone.

- **Scroll Wheel (OK Button):** Large central oval/round hybrid. Features a rubberized grip ring.
  - **Click:** Center press for selection.
  - **Scroll:** Vertical rotation for list navigation.
  - **LED Ring:** Circumferential light pipe surrounding the wheel base.
- **Directional Pad (D-Pad):** Four discrete tactile bumps surrounding the Scroll Wheel (Up, Down, Left, Right). Distinct from the wheel click.
- **Microphone Button:** Located directly below the Scroll Wheel. Concave, textured surface. Blue LED indicator embedded in the center.

5.4 Lower Middle Section: Content & Numeric Input

- **Number Pad (0-9):** 3×4 grid layout. Slightly convex keys for finger guidance.
  - **0 Key:** Double-width or standard width depending on regional variant (Standard for AKB76043105).
- **Streaming Shortcuts:** Three dedicated keys located to the right of the number pad or below it (Model dependent).
  - Netflix (Red ‘N’)
  - Disney+ (Blue Icon)
  - Amazon Prime Video (Blue Icon)
- **LIST Button:** Left of the number pad. Icon: Three horizontal lines. Opens channel list or app drawer.

5.5 Bottom Section: Color & Accessibility

- **Color Keys:** Four horizontal keys at the very bottom edge.
  - Red (A), Green (B), Yellow (C), Blue (D).
  - Textured ridges for non-visual identification.
- **Accessibility Cluster:** Located just above the color keys.
  - **Audio Description:** Icon: AD or Speaker with text.
  - **Audio Guidance:** Icon: Person with speech bubble.
  - **Magnifier:** Icon: Plus sign in lens.

5.6 Back Side

- **Battery Cover:** Sliding mechanism, bottom 40% of the rear panel.
- **Pairing Button:** Recessed circular button inside the battery compartment, near the top. Requires a pin or paperclip for actuation.
- **Label:** Regulatory markings, Model Number (AKB76043105), FCC ID, and MAC Address barcode.

5.7 LED Indicator Locations

1. **Scroll Wheel Ring:** Primary status indicator. White for active pointer, Red for pairing, Amber for low battery.
2. **Mic Button:** Pulses blue during voice listening, solid blue during processing.
3. **Top Status Light:** (If equipped) Indicates IR transmission or general system alerts.

5.8 Button Layout Diagram Description

The remote is vertically symmetrical. The **Top 20%** is reserved for Power/Volume/Channel. The **Upper Middle 25%** handles System/App navigation. The **Center 25%** is the Pointer/Selection cluster, which must be reachable by the thumb without grip adjustment. The **Lower Middle 20%** contains the numeric keypad and streaming shortcuts. The **Bottom 10%** houses the Color keys and Accessibility features.

All buttons on the front face must be flush with the bezel when unpressed, with a maximum travel distance of 0.6mm and an actuation force of 2.5N $\pm$ 0.5N. The Scroll Wheel must have 24 distinct detents per full rotation.

End of Part 1

LG AKB76043105 Magic Remote Button Function Specification

Part 2: Button Function Specifications

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6. Button Function Specifications — Part A: Navigation & Selection

This section details the functional specifications for navigation and selection controls on the LG AKB76043105 Magic Remote. These keys form the primary interaction layer for menu traversal, content selection, and pointer-based interaction. Test engineers must validate all timing thresholds, context-dependent behaviors, and mode transitions described below.

6.1 Scroll Wheel / OK Key

Field	Specification
Button Name	Scroll Wheel / OK
Chinese Name	滚轮 / 确认键
Location	Center of the directional pad; tactile center point with integrated scroll mechanism.
Short Press (<1s)	Confirms the currently highlighted selection in menus; toggles Play/Pause in video playback; enters sub-menus or launches selected applications.
Long Press (>1s)	Activates Voice Recognition / LG ThinQ AI search overlay (behavior dependent on webOS version and active service configuration).
Long Press (>3s)	In active video playback, cycles through transport states: Play → Pause → Stop. In non-playback contexts, no action.
Long Press (>5s)	Triggers Bluetooth Pairing / Re-pairing Mode. **Constraint:** Must be executed within 10 seconds of remote power-on. Remote LED enters rapid blink sequence.
Pointer Mode Behavior	Performs a Left-Click action at the current cursor coordinates. Cursor highlight follows pointer position; selection occurs upon release.
Context Dependency	**Home Dashboard:** Launches focused app/tile.**Video Player:** Toggles Play/Pause.**Settings:** Saves current parameter and exits to previous level.**Web Browser:** Navigates to focused hyperlink or submits form field.**Keyboard Input:** Confirms character entry or closes virtual keyboard.
LED Feedback	Blue LED ring surrounding the scroll wheel blinks once (duration: 100ms) upon actuation confirmation.
Response Time	⩽200ms (Bluetooth Low Energy Mode); ⩽100ms (IR Fallback Mode).
Related Settings Path	Settings > All Settings > General > Magic Remote Settings > Pointer Settings > Pointer Speed / OK Key Function.
Test Notes	1. Verify Bluetooth pairing timeout is strictly 10 seconds post-power-on.2. Test Voice Recognition activation latency.3. Validate Play/Pause/Stop cycle in video player.4. Confirm left-click accuracy in pointer mode across screen quadrants.5.

6.2 Directional Keys (Up / Down / Left / Right)

Field	Specification
Button Name	Directional Pad (D-Pad) — Up, Down, Left, Right
Chinese Name	方向键（上 / 下 / 左 / 右）
Location	Surrounding the Scroll Wheel OK key; four discrete tactile switches arranged in cardinal directions.
Short Press (<1s)	Moves focus cursor one unit in the corresponding direction within menus, grids, and UI lists.
Long Press (>1s)	Initiates continuous navigation (auto-repeat) in the pressed direction. Repeat rate: 10 items/sec initially, ramping to 20 items/sec after 2s.
Long Press (>3s)	In video playback, triggers Seek Forward/Backward by 10-second increments per repeat tick.
Long Press (>5s)	No special function; continuous navigation/seek behavior persists.
Pointer Mode Behavior	**Immediate Mode Switch:** Pressing any D-Pad key instantly hides the on-screen pointer/cursor and switches the remote to traditional D-Pad navigation mode. Pointer reactivation requires a physical shake of the remote or a press of the OK wheel.
Context Dependency	**Home Dashboard:** Moves focus between app icons/tiles.**Settings:** Navigates between setting categories and parameters.**Video Player:** Navigates between subtitle/audio tracks when OSD is visible.**Web Browser:** Moves focus between hyperlinks, form fields, and scrollable regions.**Gallery:** Scrolls content or navigates between images.
LED Feedback	None. Directional keys do not trigger LED feedback.
Response Time	⩽150ms (Bluetooth Mode); ⩽80ms (IR Mode).
Related Settings Path	Settings > All Settings > General > Magic Remote Settings > Pointer Speed / D-Pad Repeat Rate.
Test Notes	1. Verify pointer hide/show transition is seamless (no flicker, no lag).2. Test rapid D-Pad presses for correct repeat rate ramp-up.3. Validate seek granularity in video player during long press.4. Confirm pointer reactivation via shake

6.3 Back / Backward Key

Field	Specification
Button Name	Back / Backward
Chinese Name	返回键
Location	Upper section, right side of Home key; distinct tactile profile.
Short Press (<1s)	Navigates to previous menu level; exits current application; closes OSD, pop-up menus, or floating panels.
Long Press (>1s)	In Web Browser, navigates to previous page in browser history stack.
Long Press (>3s)	In Video Player, restarts current playback from the beginning (00:00:00).
Long Press (>5s)	No special function; behavior continues as per >3s definition.
Pointer Mode Behavior	Functions identically to non-pointer mode. No pointer-specific override or dismissal behavior.
Context Dependency	**Home Dashboard:** No action (already at root level).**Application:** Closes app and returns to Home Dashboard or previous app.**Web Browser:** Goes to previous page in history.**Settings:** Returns to previous settings category.**Live TV:** Closes channel banner/OSD if visible; otherwise no action.
LED Feedback	None.
Response Time	⩽200ms.
Related Settings Path	Settings > All Settings > General > Back Button Settings > Back Key Behavior.
Test Notes	1. Verify behavior at each UI depth level (min 5 levels deep).2. Test nested app navigation and proper state restoration.3. Validate browser history stack integrity after multiple Back presses.4. Confirm video restart resets progress bar and metadata.5. Test Back key during active voice recognition session (should cancel).

7. Button Function Specifications — Part B: System & Menu

This section covers system-level and menu-access keys. These controls manage global state, application lifecycle, settings access, and information overlays. Special attention must be paid to combination key sequences and long-press timing windows.

7.1 Home / Smart Home Key

Field	Specification
Button Name	Home / Smart Home
Chinese Name	主页 / 智能主页键
Location	Upper section, center; prominent tactile button with house icon.
Short Press (<1s)	Opens the full-screen Home Dashboard (webOS Launcher). Pauses active media if applicable.
Long Press (>1s)	Opens App Switcher / Recent Apps Panel, displaying the last 5 active applications in MRU order.
Long Press (>3s)	Opens Home Dashboard with automatic focus placed on the Search Bar input field.
Long Press (>5s)	**Combination Key:** Requires simultaneous long press of Home + Back for 5 seconds. Triggers Bluetooth Deregistration (Pairing Reset). Remote LED blinks Red 3 times to confirm.
Pointer Mode Behavior	Pressing Home immediately switches from Pointer Mode to D-Pad Mode and opens Home Dashboard. Pointer is deactivated.
Context Dependency	**Any Screen:** Opens Home Dashboard.**Home Dashboard:** Opens Recent Apps Panel.**Video Player:** Opens Home Dashboard; pauses video.**Settings:** Opens Home Dashboard; auto-saves unsaved changes.**Live TV:** Opens Home Dashboard; mutes audio if configured.
LED Feedback	Blue LED blinks once on short press. Red LED blinks 3 times on 5-second combination hold (deregister confirmation).
Response Time	⩽300ms.
Related Settings Path	Settings > All Settings > General > Magic Remote Settings > Home Button > Home Key Function.
Test Notes	1. Verify 5-second hold timing precisely using stopwatch/tooling.2. Test deregister and re-pair flow end-to-end.3. Validate Recent Apps MRU ordering after multiple app switches.4. Confirm auto-save behavior in Settings before Home exit.5. Test Home key during firmware update (should be blocked).

7.2 Settings / Q.Settings Key

Field	Specification
Button Name	Settings / Q.Settings
Chinese Name	设置 / 快捷设置键
Location	Upper section, right side; gear icon.
Short Press (<1s)	Opens Quick Settings floating panel overlay. Displays: Picture Mode, Sound Mode, Input Source, Wi-Fi Status, Accessibility Shortcuts.
Long Press (>1s)	Opens All Settings (full hierarchical settings menu tree).
Long Press (>3s)	When Quick Settings panel is already open, toggles Picture Mode cyclically: Standard → Expert → Calibrated → HDR Effect → Standard.
Long Press (>5s)	No special function.
Pointer Mode Behavior	Quick Settings panel appears as a floating overlay. Pointer remains active and can be used to click individual settings items without mode switching.
Context Dependency	**Home:** Opens Quick Settings overlay.**Video Player:** Opens Quick Settings with video-specific options (Picture Mode, Sound Mode, Subtitle, Audio Track).**Settings Menu:** Opens All Settings directly (bypasses Quick Settings).**Web Browser:** Opens Quick Settings with browser-specific options (Zoom, Text Size).**Live TV:** Opens Quick Settings with channel/audio options.
LED Feedback	Blue LED blinks once on press.
Response Time	⩽250ms.
Related Settings Path	Settings > All Settings > (Various Sub-menus); Quick Settings items are configurable via Settings > General > Quick Settings.
Test Notes	1. Verify Quick Settings vs. All Settings entry point distinction.2. Test context-specific option availability in each app.3. Validate Picture Mode cycle order and persistence.4. Confirm pointer interaction with Quick Settings items.5. Test Quick Settings overlay dismissal via Back/Exit/OK.

7.3 Exit Key

Field	Specification
Button Name	Exit
Chinese Name	退出键
Location	Upper section, far right; distinct “X” or arrow-out icon.
Short Press (<1s)	Clears all floating windows, OSD menus, and overlays. Returns directly to current TV channel or video playback screen.
Long Press (>1s)	Closes the current foreground application and returns to the last active screen (Home or previous app).
Long Press (>3s)	In Web Browser, stops loading the current page and navigates to browser homepage.
Long Press (>5s)	No special function.
Pointer Mode Behavior	Functions identically to non-pointer mode. Clears all pointer-mode overlays, context menus, and floating panels.
Context Dependency	**Any OSD/Menu:** Closes OSD and returns to content.**Application:** Exits app to last active screen.**TV Channel:** No action (already at content level).**Web Browser:** Stops page load; goes to browser homepage.**PiP Mode:** Closes PiP window first, then content on second press.
LED Feedback	None.
Response Time	≤200ms.
Related Settings Path	N/A (System-level hard-coded behavior).
Test Notes	1. Test from every possible UI state (min 20 states).2. Verify no orphaned overlays remain after Exit.3. Validate app close behavior and memory cleanup.4. Test browser stop-load during active network request.5. Confirm PiP close sequence priority.

7.4 Info / Information Key

Field	Specification
Button Name	Info / Information
Chinese Name	信息键
Location	Upper section, below Settings key; “i” icon.
Short Press (<1s)	Displays detailed information overlay for current content: Resolution, Aspect Ratio, Signal Strength, Program Guide (EPG), Audio Format, Subtitle Status, Bitrate.
Long Press (>1s)	Toggles between Full Info Overlay (detailed) and Compact Info Overlay (minimal).
Long Press (>3s)	In Live TV, displays the full Electronic Program Guide (EPG) for the current channel, focused on current time slot.
Long Press (>5s)	No special function.
Pointer Mode Behavior	Info overlay can be dismissed by clicking anywhere outside the overlay bounds using the pointer. Overlay remains active during pointer movement.
Context Dependency	**Live TV:** Shows channel info, signal strength, program details, next program.**HDMI Input:** Shows input source details, connected device info, supported resolutions, HDR format.**USB/Streaming:** Shows file info, codec details, playback stats, container format.**Web Browser:** Shows page URL, loading status, connection security, page size.**Gallery:** Shows image/video metadata (EXIF, resolution, date).
LED Feedback	Blue LED blinks once on press.
Response Time	⩽250ms.
Related Settings Path	Settings > All Settings > Channel > Program Guide > Info Display Settings.
Test Notes	1. Verify info content accuracy for each input source.2. Test overlay dismissal via pointer click outside bounds.3. Validate Full/Compact toggle persistence.4. Confirm EPG launch timing and focus position.5. Test Info key during active PiP (should show main content info).

7.5 Guide / List Key

Field	Specification
Button Name	Guide / List
Chinese Name	节目指南 / 列表键
Location	Lower middle section, near number keys; list/grid icon.
Short Press (<1s)	Opens the full Electronic Program Guide (EPG) / Channel List in full-screen mode.
Long Press (>1s)	Opens the Channel List in compact sidebar overlay mode.
Long Press (>3s)	Opens the Guide with the current channel pre-selected and highlighted at current time position.
Long Press (>5s)	No special function.
Pointer Mode Behavior	Guide/Channel List can be navigated using the pointer to select channels, time slots, or categories. Scroll wheel also functional for vertical navigation.
Context Dependency	**Live TV (Antenna/Cable):** Opens EPG with current time indicator and channel grid.**Satellite/IPTV:** Opens provider-specific guide interface.**Streaming App:** Opens content library/guide for the active streaming application.**Non-TV Input:** Opens last active TV guide or displays “No Guide Available” message.
LED Feedback	Blue LED blinks once on press.
Response Time	≤300ms.
Related Settings Path	Settings > All Settings > Channel > Program Guide > Guide Settings > Guide Display Mode.
Test Notes	1. Test with various TV sources (Antenna, Cable, Satellite, IPTV).2. Verify guide data accuracy and time synchronization.3. Validate compact vs. full-screen mode transition.4. Test pointer navigation within guide grid.5. Confirm current channel highlight accuracy after channel change.

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Part 3: Button Function Specifications

8. Button Function Specifications — Part C: Audio/Video Control

This section details the functional specifications for audio and video control keys located in the upper quadrant of the LG AKB76043105 Magic Remote. These keys are critical for primary user interaction and must adhere to strict latency and context-awareness requirements.

8.1 Volume Up Key

Field	Specification
Button Name	Volume Up
Chinese Name	音量加
Location	Top section, right side of the directional cluster
Short Press (<1s)	Increases system volume by exactly 1 level on the 0-100 scale. Triggers immediate OSD volume bar update.
Long Press (>1s)	Initiates continuous volume ramp-up. Repeats increment every 0.5s until key release. Ramp rate may accelerate after 3s continuous hold.
Long Press (>3s)	In specific Accessibility modes, activates Voice Guidance volume adjustment overlay. Standard volume ramp continues in non-accessibility modes.
Long Press (>5s)	No special function defined. Continuous ramp persists until release or max volume (100) reached.
Pointer Mode Behavior	Functions identically to standard mode. Pointer position does not affect volume logic.
Context Dependency	**TV Speaker:** Adjusts internal speaker gain.**HDMI-CEC (Soundbar/AVR):** Sends CEC `User Control Pressed [Volume Up]` to external device; TV volume OSD reflects external device state.**Bluetooth Audio:** Adjusts digital pre-gain or sends AVRCP Volume Up command.**Headphone Mode:** Adjusts headphone output level independently of TV speaker.**Muted State:** First press unmutes audio and restores volume to minimum 30% (configurable), subsequent presses increase normally.
LED Feedback	No LED feedback. Visual confirmation via OSD only.
Response Time	⩽80ms (IR transmission); ⩽150ms (Bluetooth RF); ⩽200ms (OSD render latency).
Related Settings Path	Settings > All Settings > Sound > Sound Output; Settings > All Settings > Accessibility > Voice Guidance
~~**Test Notes**~~	~~1. Verify step size consistency (typically~~ 4% per press in UI, 1 unit in backend).2. Test CEC volume sync: ensure TV OSD updates when external amp changes volume via its own remote.3. Verify

8.2 Volume Down Key

Field	Specification
Button Name	Volume Down
Chinese Name	音量减
Location	Top section, directly below Volume Up key
Short Press (<1s)	Decreases system volume by exactly 1 level on the 0-100 scale. Triggers immediate OSD volume bar update.
Long Press (>1s)	Initiates continuous volume ramp-down. Repeats decrement every 0.5s until key release.
Long Press (>3s)	In specific Accessibility modes, activates Voice Guidance volume adjustment overlay. Standard volume ramp continues in non-accessibility modes.
Long Press (>5s)	No special function defined. Continuous ramp persists until release or min volume (0) reached.
Pointer Mode Behavior	Functions identically to standard mode.
Context Dependency	**TV Speaker:** Decreases internal speaker gain.**HDMI-CEC:** Sends CEC `User Control Pressed [Volume Down]`.**Bluetooth Audio:** Decreases digital pre-gain or sends AVRCP Volume Down.**Headphone Mode:** Decreases headphone output level.**At Volume 0:** Automatically engages Mute state. OSD displays “Muted” icon.
LED Feedback	No LED feedback.
Response Time	⩽80ms (IR); ⩽150ms (Bluetooth); ⩽200ms (OSD render).
Related Settings Path	Settings > All Settings > Sound > Sound Output
Test Notes	1. Verify mute trigger at volume 0.2. Test ramp-down acceleration curve matches ramp-up.3. Ensure CEC volume down does not desync from TV OSD.4. Verify headphone volume does not affect TV speaker volume when both are active (if supported).

8.3 Mute Key

Field	Specification
Button Name	Mute
Chinese Name	静音
Location	Top section, center of directional cluster
Short Press (<1s)	Toggles Mute state ON/OFF. ON: Audio path silenced, “Muted” OSD displayed. OFF: Audio restored to pre-mute level.
Long Press (>1s)	Toggles Audio Description (AD) / Secondary Audio Program (SAP) on supported models. If AD unavailable, no action.
Long Press (>3s)	Cycles through audio tracks: Main Audio → Secondary Audio → Audio Description (model/market dependent).
Long Press (>5s)	No special function.
Pointer Mode Behavior	Mute OSD overlay appears as standard layer. Pointer remains active for navigation while muted.
Context Dependency	**TV Speaker:** Silences internal speakers.**HDMI-CEC:** Sends CEC `User Control Pressed [Mute]` to external audio device.**Bluetooth:** Silences Bluetooth stream; TV speaker may remain active based on “Sound Out” setting.**Headphone:** Silences headphone output.**Smart TV Phone App:** If active call in progress, ends call instead of muting.
LED Feedback	Red LED blinks once (50ms on, 50ms off) upon entering Mute state. No LED on Unmute.
Response Time	⩽150ms (Audio path silence); ⩽200ms (OSD update).
Related Settings Path	Settings > All Settings > Sound > Mute; Settings > All Settings > Accessibility > Audio Description
Test Notes	1. Verify pre-mute volume restoration accuracy.2. Test AD toggle in regions with broadcast AD support.3. Confirm CEC mute sync with external soundbars.4. Verify call termination priority over mute in Phone App.

8.4 Channel Up Key

Field	Specification
Button Name	Channel Up
Chinese Name	频道加
Location	Top section, left side of directional cluster
Short Press (<1s)	Switches to next channel in user-defined channel list. OSD shows channel info for 3s.
Long Press (>1s)	Initiates continuous channel scan upward. Repeat rate: 0.8s initially, accelerating to 0.3s after 3s hold.
Long Press (>3s)	Opens Channel List overlay with focus on next channel position. Scan pauses until selection or release.
Long Press (>5s)	No special function.
Pointer Mode Behavior	Identical to standard mode. Channel List overlay supports pointer selection.
Context Dependency	**Antenna/Cable:** Cycles through saved channel map.**IPTV:** Cycles through provider channel list.**Streaming App:** Triggers “Next Episode” or “Skip Ad” if supported by app.**HDMI Input:** Cycles to next HDMI port (HDMI1 → HDMI2 → … → TV).**USB/PC Mode:** Navigates to next file in media browser list.
LED Feedback	No LED feedback.
Response Time	⩽200ms (Channel switch); ⩽300ms (HDMI source switch).
Related Settings Path	Settings > All Settings > Channel > Channel List; Settings > All Settings > General > External Device Manager
Test Notes	1. Verify scan order matches user-sorted channel list.2. Test boundary behavior: Top channel + CH+ wraps to bottom channel.3. Confirm HDMI source rotation excludes disabled inputs.4. Verify app-specific “Next” behavior does not interfere with channel map.

8.5 Channel Down Key

Field	Specification
Button Name	Channel Down
Chinese Name	频道减
Location	Top section, above Channel Up key
Short Press (<1s)	Switches to previous channel in user-defined channel list. OSD shows channel info for 3s.
Long Press (>1s)	Initiates continuous channel scan downward. Repeat rate: 0.8s initially, accelerating to 0.3s after 3s hold.
Long Press (>3s)	Opens Channel List overlay with focus on previous channel position.
Long Press (>5s)	No special function.
Pointer Mode Behavior	Identical to standard mode.
Context Dependency	Same as Channel Up, but in reverse direction. HDMI rotation: HDMI4 → HDMI3 → ... → TV.
LED Feedback	No LED feedback.
Response Time	⩽200ms (Channel switch); ⩽300ms (HDMI source switch).
Related Settings Path	Settings > All Settings > Channel > Channel List
Test Notes	1. Verify reverse scan order.2. Test boundary behavior: Bottom channel + CH- wraps to top channel.3. Confirm acceleration curve matches Channel Up.

8.6 Input / Source Key

Field	Specification
Button Name	Input / Source
Chinese Name	输入/信号源
Location	Upper section, below Exit key
Short Press (<1s)	Opens Input Source selector overlay (Quick Picker). Displays icons for TV, HDMI 1-4, USB, etc.
Long Press (>1s)	Opens full-screen Input Source Manager with device names, icons, and edit capabilities.
Long Press (>3s)	Directly switches to next available HDMI input in rotation (skips TV/USB).
Long Press (>5s)	No special function.
Pointer Mode Behavior	Overlay supports pointer hover-highlight and click-select. Full-screen manager supports drag-reorder.
Context Dependency	**From TV:** Shows all external inputs.**From HDMI:** Shows all inputs including TV.**From USB:** Shows all inputs including TV and HDMI.**From Home:** Shows all inputs.**Disabled Inputs:** Hidden from list unless “Show Disabled” setting enabled.
LED Feedback	Blue LED blinks once (100ms) on press.
Response Time	⩽250ms (Overlay render); ⩽500ms (Source switch handshake).
Related Settings Path	Settings > All Settings > General > External Device Manager > Input Device Manager
Test Notes	1. Verify all connected devices appear with correct labels.2. Test custom input naming persistence.3. Verify HDMI hot-plug detection updates list immediately.4. Test long-press direct switch skips disabled HDMI ports.

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9. Button Function Specifications — Part D: Smart & Shortcut Keys

This section covers AI interaction, voice control, and dedicated application shortcut keys. These functions rely heavily on software state and network connectivity.

9.1 Voice / Microphone Key

Field	Specification
Button Name	Voice / Microphone
Chinese Name	语音/麦克风
Location	Below Scroll Wheel OK key, center navigation cluster
Short Press (<1s)	No function. Key requires minimum 200ms hold to activate. Prevents accidental triggers.
Long Press (>1s)	Activates LG ThinQ AI Voice Assistant. LED ring glows Blue. User speaks command. Release button to submit.
Long Press (>3s)	Enters Continuous Listening Mode (Hands-Free). LED pulses Blue. System listens for wake word or commands without holding. Timeout: 10s silence.
Long Press (>5s)	No special function. Continuous mode persists until timeout or explicit “Stop” command.
Pointer Mode Behavior	Voice results appear as OSD overlay. Pointer can select suggested actions/ cards from voice response.
Context Dependency	**Home Screen:** “Open [App]” launches app.**Live TV:** “Change to channel 5” switches channel.**Settings:** “Turn on Bluetooth” toggles setting.**Video Player:** “Subtitles on”, “Volume up” executes playback control.**Browser:** “Search for [query]” initiates web search.**Global:** “What’s the weather?” shows weather widget.**IoT:** “Turn on [Device]” controls linked smart home devices.
LED Feedback	**Blue Solid:** Listening/Processing.**Blue Pulse:** Continuous Mode Active.**Red Blink:** Recognition Error/No Network.**Off:** Idle.
Response Time	Activation: ⩽500ms; Command Execution: ⩽2s (Network dependent).
Related Settings Path	Settings > All Settings > General > Magic Remote Settings > Voice Recognition; Settings > All Settings > General > AI Service
Test Notes	1. Test in high-noise environments (60dB+).2. Verify offline fallback for basic commands (Volume, Mute).3. Test continuous mode timeout accuracy.4. Verify LED state transitions match audio feedback.

9.2 Netflix Key

Field	Specification
Button Name	Netflix
Chinese Name	Netflix 快捷键
Location	Lower middle section, left shortcut key
Short Press (<1s)	Launches Netflix App. If backgrounded, resumes foreground. If not installed, opens LG Content Store to Netflix page.
Long Press (>1s)	Launches Netflix with focus on Search input field.
Long Press (>3s)	No special function.
Long Press (>5s)	No special function.
Pointer Mode Behavior	App opens in full-screen. Pointer navigation enabled within Netflix UI.
Context Dependency	**Home:** Standard launch.**Other App:** App switcher behavior (preserves state).**Netflix:** Opens Search overlay.**No Install:** Content Store deep link.
LED Feedback	Blue LED blinks once on press.
Response Time	⩽500ms (App launch trigger); App load time variable.
Related Settings Path	N/A (System App)
Test Notes	1. Test cold launch vs. warm resume.2. Verify deep link to search works.3. Test behavior when Netflix is uninstalled/ disabled.

9.3 Disney+ Key

Field	Specification
Button Name	Disney+
Chinese Name	Disney+ 快捷键
Location	Lower middle section, center shortcut key
Short Press (<1s)	Launches Disney+ App. Resumes if backgrounded. Redirects to Store if missing.
Long Press (>1s)	Launches Disney+ with focus on Search.
Long Press (>3s)	No special function.
Long Press (>5s)	No special function.
Pointer Mode Behavior	Full-screen launch; pointer navigation supported.
Context Dependency	Identical logic to Netflix Key.
LED Feedback	Blue LED blinks once on press.
Response Time	⩽500ms.
Related Settings Path	N/A
Test Notes	1. Verify app switching state preservation.2. Test search focus long-press.

9.4 Amazon Prime Video Key

Field	Specification
Button Name	Amazon Prime Video
Chinese Name	Prime Video 快捷键
Location	Lower middle section, right shortcut key
Short Press (<1s)	Launches Amazon Prime Video App. Resumes if backgrounded.
Long Press (>1s)	Launches Prime Video with focus on Search.
Long Press (>3s)	No special function.
Long Press (>5s)	No special function.
Pointer Mode Behavior	Full-screen launch; pointer navigation supported.
Context Dependency	Identical logic to Netflix Key.
LED Feedback	Blue LED blinks once on press.
Response Time	⩽500ms.
Related Settings Path	N/A
Test Notes	1. Verify regional app variant launches correctly.2. Test search focus.

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10. Button Function Specifications — Part E: Number Keys & Input

This section defines the alphanumeric input matrix and utility keys. Note the dual-function nature of number keys (Short Press = Input, Long Press = Shortcut).

10.1 Number Keys (0-9)

Field	Specification
Button Name	Number Keys (0-9)
Chinese Name	数字键 (0-9)
Location	Lower middle section, 3x4 grid layout
Short Press (<1s)	**TV Mode:** Direct Channel Entry (e.g., 1-2-3 → Ch 123).**App/Input:** Inputs digit into active text field.**Phone App:** DTMF tone generation.
Long Press (>1s)	**Key 1:** Open Favorites List.**Key 2:** Open Quick Settings Panel.**Key 3:** Open Guide/EPG.**Key 4:** Open Input Source Selector.**Key 5:** Open Info/Program Details.**Key 6:** Open Channel List.**Key 7:** Toggle Teletext (if available).**Key 8:** Open Accessibility Shortcuts.**Key 9:** Open Recent Apps Panel.**Key 0:** Open On-Screen Number Pad/Keyboard.
Long Press (>3s)	No special function for most keys. Key 7 may toggle Teletext language if held.
Long Press (>5s)	No special function.
Pointer Mode Behavior	Short press input identical. Long-press shortcuts open overlays navigable via pointer.
Context Dependency	**TV:** Channel tuning.**Browser:** URL/Search bar input.**Settings:** Numeric value adjustment.**Phone:** Dial pad input.**Gaming:** May map to D-Pad/Buttons if “Game Optimizer” remapping active.
LED Feedback	No LED feedback.
Response Time	⩽150ms (Input); ⩽300ms (Shortcut Launch).
Related Settings Path	Settings > All Settings > General > Magic Remote Settings > Button Shortcut
Test Notes	1. Test multi-digit channel entry timing (timeout between digits).2. Verify all 10 long-press shortcuts function.3. Test input in password fields (masking).4. Verify Key 0 opens virtual keyboard, not just numpad.

10.2 LIST / Channel List Key

Field	Specification
Button Name	LIST / Channel List
Chinese Name	列表/频道列表
Location	Lower middle section, above number keys
Short Press (<1s)	Opens saved Channel List / Program Guide overlay.
Long Press (>1s)	Opens Channel List with current channel pre-highlighted.
Long Press (>3s)	Opens Channel List in Full-Screen Mode (replaces video).
Long Press (>5s)	No special function.
Pointer Mode Behavior	List supports pointer scroll and click-select.
Context Dependency	**Live TV:** Standard channel list.**Cable/Sat:** Provider guide.**Streaming App:** Opens content library/category list.**HDMI:** No function or opens external device menu (CEC).
LED Feedback	Blue LED blinks once on press.
Response Time	≤300ms.
Related Settings Path	Settings > All Settings > Channel > Channel List
Test Notes	1. Verify sort order (User vs. Provider).2. Test full-screen mode toggle.3. Verify current channel highlight accuracy.

10.3 Red / Green / Yellow / Blue Button Keys

Field	Specification
Button Name	Red / Green / Yellow / Blue
Chinese Name	红/绿/黄/蓝 彩色键
Location	Bottom section, horizontal row
Short Press (<1s)	**HbbTV/Teletext:** Executes on-screen colored action.**Netflix/Prime:** App-specific shortcuts (e.g., Red=My List).**TV Mode:** Contextual function (see Long Press defaults).
Long Press (>1s)	**Red:** Open Recording/Schedule

LG AKB76043105 Magic Remote Button Function Specification

Part 4: Button Function Specifications — Part F: Accessibility & Special Keys, Pointer Mode Detailed Specification, Combination Keys & Gesture Specification

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11. Button Function Specifications — Part F: Accessibility & Special Keys

This section defines the functional behavior, timing parameters, and system interactions for accessibility-focused and special-purpose keys on the LG AKB76043105 Magic Remote. These keys are designed to comply with WCAG 2.1 guidelines and provide immediate access to system-level assistive technologies.

11.1 Accessibility Key (Wheelchair Icon)

Field	Specification
Button Name	Accessibility / Accessibility Shortcuts
Chinese Name	无障碍 / 无障碍快捷方式
Location	Bottom section, left side of the directional cluster
Short Press (<1s)	Opens the **Accessibility Shortcuts Overlay** (floating panel). Displays toggles for: Voice Guide On/Off, High Contrast, Large Text, Color Filters, Mono Audio, and Slow Remote.
Long Press (>1s)	Navigates directly to the full **Accessibility Settings** menu path: `All Settings > General > Accessibility`.
Long Press (>3s)	Immediately toggles **Voice Guide (Audio Description)** state (On ↔ Off) without opening any UI overlay.
Long Press (>5s)	No special function assigned.
Pointer Mode Behavior	When the pointer is active, the Accessibility Shortcuts Overlay supports direct point-and-click selection of individual toggles. The pointer must be within the bounding box of the toggle to activate.
Context Dependency	**Global:** Opens Accessibility Shortcuts Overlay from any active input or app.**Settings Menu:** Opens full Accessibility Settings menu.**Voice Guide Active:** The toggle in the overlay reflects current state and turns it Off upon selection.
LED Feedback	**Blue LED:** Blinks once on short press.**Green LED:** Solid glow indicates Voice Guide is currently active.
Response Time	⩽250ms for overlay launch; ⩽100ms for Voice Guide toggle.
Related Settings Path	`Settings > All Settings > General > Accessibility`
Test Notes	1. Verify overlay dismissal via OK button, Back button, or pointer click outside the panel.2. Confirm Voice Guide toggle persists across power cycles.3. Test overlay behavior during active video ~~playback (must not pause content).~~

11.2 Audio Guidance Key (Speaker Icon)

Field	Specification
Button Name	Audio Guidance / Voice Guide
Chinese Name	语音引导 / 语音指南
Location	Bottom section, center-left, adjacent to Accessibility Key
Short Press (<1s)	Toggles **Voice Guide (Text-to-Speech)** On/Off. When On, the TV audibly announces focused UI elements, menu navigation, and system events.
Long Press (>1s)	Opens **Voice Guide Configuration Menu**: Speed, Pitch, Volume, and Language selection.
Long Press (>3s)	Model-dependent: Toggles between **Voice Guide** and **Audio Description** (secondary audio track for visually impaired content).
Long Press (>5s)	No special function assigned.
Pointer Mode Behavior	Voice Guide automatically reads out the UI element currently under the pointer focus. Reading is triggered upon pointer entry into a focusable zone.
Context Dependency	**Global:** Toggles Voice Guide On/Off.**Settings:** Opens Voice Guide configuration.**Video Playback:** May toggle between Main Audio and Descriptive Audio track if available.
LED Feedback	**Green LED:** Solid glow when Voice Guide is active.**No LED:** When Voice Guide is disabled.
Response Time	⩽200ms for state toggle; ⩽300ms for settings menu launch.
Related Settings Path	`Settings > All Settings > Sound > Audio Output > Voice Guide`
Test Notes	1. Verify TTS reads correct menu hierarchy.2. Test speed/pitch adjustments in real-time.3. Confirm Audio Description track switching during broadcast playback.

11.3 Zoom / Magnifier Key

Field	Specification
Button Name	Zoom / Magnifier
Chinese Name	缩放 / 放大镜
Location	Bottom section, center-right
Short Press (<1s)	Activates **Screen Zoom (Magnifier Mode)**. Default magnification: 150%. The pointer/cursor becomes the viewport controller for panning.
Long Press (>1s)	Cycles through predefined zoom levels: **100% (Off)** → **150%** → **200%** → **300%** → **150%**.
Long Press (>3s)	Sets zoom to **Maximum (300%)** and centers the viewport on the current pointer coordinates.
Long Press (>5s)	No special function assigned.
Pointer Mode Behavior	**In Zoom Mode:** Pointer controls viewport position. Scroll Wheel adjusts zoom level incrementally. Directional Keys pan the view in discrete steps.
Context Dependency	**Home Dashboard:** Zooms the entire dashboard UI.**Settings:** Zooms the settings menu overlay.**Video Player:** Zooms the video rendering area (may be disabled during DRM-protected full-screen playback).**Web Browser:** Zooms the webpage content.
LED Feedback	**Blue LED:** Blinks once on activation.**Blue LED:** Blinks twice on zoom level change.
Response Time	⩽200ms for mode activation; ⩽100ms for zoom level transition.
Related Settings Path	`Settings > All Settings > General > Accessibility > Zoom`
Test Notes	1. Verify zoom persistence across app switches.2. Test pointer panning at 300% magnification for edge cases.3. Confirm scroll wheel zoom granularity.

11.4 Subtitle / Closed Caption Key (If Equipped)

Field	Specification
Button Name	Subtitle / CC
Chinese Name	字幕 / 隐藏式字幕
Location	Bottom section, right side (Model Dependent)
Short Press (<1s)	Toggles **Closed Caption / Subtitles** On/Off. Displays “CC On” or “CC Off” OSD indicator.
Long Press (>1s)	Opens **Subtitle Settings Overlay**: Language, Font Size, Color, Background Opacity, and Style.
Long Press (>3s)	Cycles through available **Subtitle Languages** (if multiple tracks are detected).
Long Press (>5s)	No special function assigned.
Pointer Mode Behavior	Subtitle settings overlay supports point-and-click navigation. Pointer hover previews font/color changes.
Context Dependency	**Live TV:** Toggles broadcast CC.**Streaming App:** Toggles app-native subtitles.**HDMI Input:** Toggles subtitle overlay for external source.**Settings:** Opens subtitle configuration.
LED Feedback	**Blue LED:** Blinks once when subtitles are toggled On.**No LED:** When subtitles are Off.
Response Time	⩽200ms for toggle; ⩽300ms for settings overlay.
Related Settings Path	`Settings > All Settings > Sound > Subtitle`
Test Notes	1. Test with CC, SDH, and burnt-in subtitle formats.2. Verify language cycling order matches audio track order.3. Confirm subtitle styling persists per-app.

11.5 Sleep Timer Key (If Equipped)

Field	Specification
Button Name	Sleep Timer
Chinese Name	睡眠定时器
Location	Bottom section (Model Dependent)
Short Press (<1s)	Opens **Sleep Timer Settings**. Cycles through: **Off** → **15min** → **30min** → **45min** → **60min** → **90min** → **120min**.
Long Press (>1s)	Immediately sets Sleep Timer to **30 Minutes**.
Long Press (>3s)	**Cancels** active Sleep Timer (sets to Off).
Long Press (>5s)	No special function assigned.
Pointer Mode Behavior	Sleep Timer overlay displays remaining time. Pointer can be used to adjust duration via on-screen slider.
Context Dependency	**Global:** Opens Sleep Timer settings.**Active Timer:** OSD shows remaining time.**Expiration:** TV executes power-off sequence (equivalent to Power Key long press).
LED Feedback	**Blue LED:** Blinks to indicate selected duration.**Red LED:** Blinks slowly when timer is active.
Response Time	⩽200ms for menu launch; ⩽100ms for immediate set/cancel.
Related Settings Path	`Settings > All Settings > General > Sleep Timer`
Test Notes	1. Verify all timer durations trigger power-off.2. Test cancellation during active countdown.3. Confirm power-off behavior during recording (must stop recording first).

11.6 Power / Standby Key

Field	Specification
Button Name	Power / Standby
Chinese Name	电源 / 待机
Location	Top section, far left (Red colored button)
Short Press (<1s)	Toggles TV Power State.**If On:** Initiates power-off sequence (OSD: “Power Off” → Standby).**If Standby:** Powers On TV (restores last active input/source).
Long Press (>1s)	Initiates **Immediate Power Off** (bypasses confirmation dialog).
Long Press (>3s)	**Factory Reset Prompt** (Only functional when TV is in Standby and held for >5s).
Long Press (>5s)	No special function (Ignored if TV is already Off).
Pointer Mode Behavior	Functions identically to D-Pad mode. Power-off confirmation dialog can be dismissed via pointer click within 3-second window.
Context Dependency	**Global:** Toggles power state.**Video Playback:** Prompts “Stop playback and power off?” (Yes/No).**Recording:** Prompts “Stop recording and power off?” (Yes/No).**Settings:** Same as above.
LED Feedback	**Red LED:** Solid glow when TV is On.**Red LED:** Slow blink when in Standby mode.
Response Time	⩽500ms (Power On); ⩽300ms (Power Off command sent).
Related Settings Path	`Settings > All Settings > General > Power Off Settings`
Test Notes	1. Verify power-on restores correct input.2. Test confirmation dialog timeout.3. **Safety:** Verify factory reset long-press requires TV to be in Standby to prevent accidental data loss.

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12. Pointer Mode (Point & Click) Detailed Specification

This section provides the comprehensive technical specification for the **Pointer/Point & Click Mode**, the signature interaction model of the LG AKB76043105 Magic

Remote. This mode relies on inertial sensor fusion and Bluetooth Low Energy (BLE) communication to map physical remote movement to on-screen cursor coordinates.

12.1 Pointer Activation & Deactivation Logic

Aspect	Technical Specification
Sensor Suite	3-axis Gyroscope + 3-axis Accelerometer (6-axis IMU)
Communication Protocol	Bluetooth 5.0 BLE (Low Energy)
Activation Trigger	Physical shake detection (Acceleration > 0.5G for >200ms)
Deactivation Trigger	Inactivity timeout: 3 seconds of no IMU delta
Re-activation	Shake gesture restores pointer to **last known screen coordinates**
Pointer Shape	Default: Circle with center dot.Customizable: Arrow, Circle, Crosshair, Large Circle
Pointer Size	Default: Medium.Customizable: Small, Medium, Large, Extra Large
Pointer Speed	Default: 5.Scale: 1 (Slow) to 10 (Fast)
Pointer Trail	Default: Off.Customizable: Off, Short, Medium, Long

12.2 Pointer Gestures & Action Mapping

Gesture / Action	Physical Input Description	Expected System Result
Pinch In	Pinch fingers together while pointing	**Zoom Out** (Web Browser, Gallery, Map)
Pinch Out	Spread fingers apart while pointing	**Zoom In** (Web Browser, Gallery, Map)
Rotate	Rotate wrist in circular motion	**Scroll/Rotate** object (Gallery, Map, 360 Video)
Shake	Quick back-and-forth lateral shake	**Activate/Deactivate** Pointer Mode
Tap	Quick vertical downward movement (<300ms)	**Left Click** on focused element
Hold & Point	Hold remote steady for >2 seconds	**Right-Click Context Menu** (if supported by app)
Flick	Quick swipe motion (Up/Down/Left/Right)	**Scroll/Page Turn** (Web Browser, EPG)

12.3 Scroll Wheel Behavior in Pointer Mode

Action	Expected Result
Scroll Forward	Scroll page/content **Downward**
Scroll Backward	Scroll page/content **Upward**
Press (Click)	**Left Click** on the currently focused UI element
Hold (>1s)	Open **Context Menu** (Right-click equivalent)

12.4 Pointer Mode State Machine

The Pointer Mode operates on a finite state machine to manage power, connectivity, and user intent.

```
[IDLE]
  |
  ├──(Shake Detected)──> [ACTIVE]
  |
[ACTIVE]
  |
  ├──(Stillness > 3s)──> [HIDING]
  |      |
  |      ├──(Shake Detected)──> [ACTIVE] (Restore Position)
  |      |
  |      └──(Timeout > 5s)──> [IDLE]
  |
  ├──(Direction Key Press)──> [D_PAD_MODE]
  |      |
  |      └──(OK Press)──> [ACTIVE]
  |
  ├──(BLE Disconnect)──> [IR_FALLBACK]
  |      |
  |      └──(BLE Reconnect)──> [ACTIVE]
  |
  └──(Battery < 10%)──> [WARNING]
         |
         └──(Battery Replace)──> [IDLE]
```

12.5 Pointer Mode Error Conditions & Recovery

Error Condition	Symptom	Recovery Action
Pointer Drift	Cursor moves slowly without physical remote movement	**Calibrate Pointer:** `Settings > Magic Remote > Calibrate`
Pointer Lag	Cursor movement lags >100ms behind remote movement	Check Bluetooth interference; reduce distance to TV; replace battery
Pointer Disappears	Cursor vanishes from screen unexpectedly	Shake remote; verify BLE pairing; check battery level
Pointer Jitter	Cursor shakes/vibrates in place while stationary	Check for physical obstructions; recalibrate; update firmware
Bluetooth Drop	Pointer disconnects intermittently	Check battery level; move closer to TV; remove 2.4GHz interference sources

12.6 Pointer Calibration Procedure

Step	Action	Expected Result
1	Navigate to `Settings > All Settings > General > Magic Remote > Pointer Calibration`	Calibration screen appears with 9 target circles
2	Follow on-screen instructions: Point to each target circle and click	Each target turns **Green** when acquired
3	Complete all 9 targets	Calibration saved; “Calibration Complete” message displayed
4	If calibration fails	“Calibration Failed” message; Option to **Retry** or **Exit**

---

13. Combination Keys & Gesture Specification

This section defines multi-key combinations and advanced gesture sequences that trigger system-level functions or accessibility shortcuts.

13.1 Key Combinations

Combination	Trigger Condition	Function
Volume Up + **Volume Down**	Press both simultaneously (<500ms delta)	Toggle **Mute** On/Off
Channel Up + **Channel Down**	Press both simultaneously (<500ms delta)	Open **Input Selector** Menu
Back + **Home**	Press both simultaneously (<500ms delta)	Open **Quick Settings** Panel
OK + **Back**	Press both simultaneously (<500ms delta)	**Force Close** current app (if unresponsive)
Power + **Volume Down**	Press both simultaneously (TV On)	Capture **Screenshot** (Saved to Gallery)

13.2 Advanced Gesture Sequences

Gesture Sequence	Description	Function
Double Shake	Two rapid shakes within 1 second	Toggle **Pointer Mode** On/Off (Alternative to single shake)
Shake + **Hold**	Shake and hold remote still	Open **Voice Search** (Microphone activation)
Circle Gesture	Draw a circle in air with remote	**Refresh** current page/app (Web Browser, Home)
Zig-Zag	Rapid left-right-left movement	**Undo** last action (if supported by app)

---

14. Test & Validation Notes

14.1 Accessibility Key Validation

- **Voice Guide Toggle:** Verify state persistence across power cycles and input switches.
- **Overlay Dismissal:** Ensure overlay closes on Back, OK, and pointer click outside bounds.
- **Conflict Check:** Verify Accessibility Key does not trigger during active video recording.

14.2 Pointer Mode Validation

- **Drift Test:** Leave remote stationary for 60 seconds; cursor must not drift >5% of screen width.
- **Latency Test:** Measure time from physical movement to on-screen update; must be <30ms at 1m distance.
- **Battery Test:** Verify [WARNING] state triggers at <10% battery; verify pointer disables at <5%.

- **Interference Test:** Operate remote with active 2.4GHz Wi-Fi and Bluetooth devices within 1m; pointer must remain stable.

14.3 Combination Key Validation

- **Simultaneity:** Verify key press delta <500ms is required; sequential presses must not trigger combination.
- **Safety:** Verify **Force Close** (OK + Back) does not corrupt system apps or settings.
- **Screenshot:** Verify **Power** + **Volume Down** saves image to correct gallery folder and displays OSD confirmation.

End of Part 4: Accessibility, Pointer Mode, and Combination Keys Specification.

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Part 5: Advanced Control, System Integration, and Compliance

13. Combination Keys & Gesture Specification

This section defines all multi-key combination presses, hold durations, and their corresponding system-level functions. These combinations are critical for service diagnostics, factory resets, and mode switching. Test engineers must verify hold timers using a stopwatch or automated test rig to ensure the remote firmware correctly distinguishes between short presses, long holds, and combination holds.

Combo Key	Keys	Hold Duration	Expected Result	Context Dependency	LED Feedback	Test Notes
Home + Back	Home + Back	5 seconds	Deregister remote from TV (Bluetooth pairing removed). Remote enters pairing mode immediately after deregistration.	From any screen	Red LED blinks 3 times	Test re-pairing after deregistration; verify only this specific remote is unpaired (not all paired devices); verify TV OSD confirms “Remote Unpaired”.
Home + Guide	Home + Guide	5 seconds	Factory reset remote firmware configuration (non-volatile memory reset). Does not affect TV user data.	From any screen	Orange LED blinks continuously for 3s	Verify data is not deleted from TV; only remote settings (macros, learned keys) are reset; verify remote reboots.
Volume Up + Down	Volume Up + Down	5 seconds	Mute all audio output and enter Service Menu (advanced diagnostics).	From any screen	No LED	**WARNING:** Service Menu access requires caution; document for QA only; verify password prompt if applicable; verify exit method (Power Off).
Back + Left	Back + Left	3 seconds	Open IR Learning Mode (Macro Mode)	From any screen (Live TV)	Red LED blinks 2 rapidly then solid Red	**WARNING:**

Gesture Recognition Specification:

Gesture	Input Device	Threshold	Expected Result	Latency Requirement
Point & Click	Gyroscope + Optical Sensor	Cursor speed > 500px/s	Left Click	< 40ms
Flick Left	Gyroscope	Angular velocity > 300°/s	Back / Previous	< 60ms
Flick Right	Gyroscope	Angular velocity > 300°/s	Forward / Next	< 60ms
Shake	Accelerometer	3-axis delta > 2G within 200ms	Cancel / Return to Live TV	< 80ms
Tilt Up	Gyroscope	Pitch > 45° for > 1s	Volume Up (Continuous)	< 100ms
Tilt Down	Gyroscope	Pitch < −45° for > 1s	Volume Down (Continuous)	< 100ms

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14. Context-Dependent Button Behavior Matrix

This matrix documents the state-dependent logic for physical buttons. The Magic Remote acts as a universal controller, and button mapping changes dynamically based on the active application, input source, and TV power state.

14.1 Behavior Matrix by TV State

Button	TV Off (Standby)	TV On - Home	TV On - App	TV On - Settings	TV On - Video Player	TV On - Browser
Power	Power On (IR/BT)	—	—	—	—	—
Volume Up	Mute toggle (Standby)	+1 Volume	+1 Volume	+1 Volume	+1 Volume	+1 Volume
Volume Down	Mute toggle (Standby)	−1 Volume	−1 Volume	−1 Volume	−1 Volume	−1 Volume
Mute	Mute toggle	Toggle Mute	Toggle Mute	Toggle Mute	Toggle Mute	Toggle Mute
Channel Up	—	—	—	—	Next Chapter/ Track	—
Channel Down	—	—	—	—	Prev Chapter/ Track	—
Input	—	Open Input List	Switch Source	—	—	—
Home	—	Open Home Dashboard	Open Home Dashboard	Open Home Dashboard	Open Home Dashboard	Open Home Dashboard
Back	—	—	Exit App / Go Back	Back / Close Menu	—	Back / Close Tab
Settings	—	Open Quick Settings	Open Quick Settings	Back / Close	Open Quick Settings	Open Quick Settings
Exit	—	—	Exit App	Close Overlay	Stop Playback	Close Overlay
Info	—	—	—	—	Show Media Info	Show Page URL/Title
Guide	—	—	—	—	Open EPG / Timeline	—
Voice	—	Activate Voice	Activate Voice	Activate Voice	Activate Voice	Activate Voice
Netflix	—	Open Netflix App	Open Netflix App	Open Netflix App	Open Netflix App	Open Netflix App
OK / Select	—	Select Tile	Select / Play	Select Option	Pause / Resume	Click Link

14.2 Behavior Matrix by Input Source

Button	TV (Antenna/ Cable)	HDMI 1 (STB)	HDMI 2 (Console)	USB Media	Bluetooth Audio	Web Browser
Channel Up	Next Channel	Vol Up (Pass-through)	—	Next Track	Next Track	Scroll Up
Channel Down	Prev Channel	Vol Down (Pass-through)	—	Prev Track	Prev Track	Scroll Down
Input	Open Source List	Open Source List	Open Source List	Open Source List	Open Source List	Open Source List
Aspect Ratio	Cycle Ratios	Cycle Ratios	Cycle Ratios	Cycle Ratios	—	Zoom In/ Out
Subtitle	Toggle CC	Toggle CC	Toggle CC	Toggle Subtitles	—	Toggle Captions
Sleep Timer	Set Timer	Set Timer	Set Timer	Set Timer	Set Timer	Set Timer
Color Button (Red)	Teletext / Red Button	STB Red Button	—	—	—	Refresh Page
Color Button (Green)	Teletext / Green Button	STB Green Button	—	—	—	Back
Color Button (Yellow)	Teletext / Yellow Button	STB Yellow Button	—	—	—	Favorites
Color Button (Blue)	Teletext / Blue Button	STB Blue Button	—	—	—	Home

15. Voice Command Specification

15.1 Voice Command Categories

Category	Example Commands	Expected Result	Latency
Channel Control	“Change channel to 5”	Switch to channel 5	< 1.5s
	“Next channel”	Switch to next channel	< 1.0s
	“Go back to previous channel”	Return to previous channel	< 1.0s
Volume Control	“Volume up”	Increase volume by 3 levels	< 0.8s
	“Volume down”	Decrease volume by 3 levels	< 0.8s
	“Set volume to 50”	Set volume to 50	< 1.2s
	“Mute” / “Unmute”	Toggle mute state	< 0.5s
App Control	“Open Netflix”	Launch Netflix app	< 2.0s
	“Open settings”	Open Quick Settings panel	< 1.0s
	“Go home”	Return to Home Dashboard	< 1.0s
	“Exit”	Exit current app / Close overlay	< 0.8s
Playback Control	“Pause” / “Play” / “Stop”	Control video playback state	< 0.5s
	“Rewind” / “Fast forward”	Control playback speed (2x, 4x)	< 0.8s
	“Seek to 5:00”	Jump to 5 minutes timestamp	< 1.5s
Search	“Search for movies”	Open search with movie results	< 2.5s
	“What’s on channel 7?”	Show program info for channel 7	< 2.0s
	“Who directed Inception?”	Show search results/knowledge graph	< 3.0s
Smart Home	“Turn on the lights”	Send command to connected smart home device	< 2.0s
	“Set temperature to 22 degrees”	Control connected thermostat	< 2.0s
System	“Turn off the TV”	Power off TV (Standby)	< 1.0s
	“Take a screenshot”	Capture screen and save to USB	< 1.5s
	“What is the weather?”	Show weather overlay	< 2.0s
	“Bluetooth settings”	Open Bluetooth settings menu	< 2.0s

15.2 Voice Command Technical Parameters

Parameter	Specification	Test Method
Wake Word	“Hi LG” (always-on listening, configurable in settings)	Verify wake word detection in 65dB ambient noise
Voice Engine	LG ThinQ AI Cloud-based (with offline fallback)	Verify cloud handshake; verify offline IR fallback
Supported Languages	English, Korean, Japanese, German, French, Spanish, Italian, Portuguese, Chinese (Simplified/Traditional)	Verify language switch in settings; verify accent recognition
Response Time	⩽2 seconds (network dependent), ⩽500ms (local command)	Measure time from mic LED on to action execution
Offline Fallback	Basic commands (volume, channel, power, mute) work offline via IR	Disconnect Wi-Fi; verify basic commands still function
Noise Tolerance	Functional in environments up to 65dB SPL	Play pink noise at 65dB; verify command success rate > 80%
Voice Recognition Accuracy	⩾95% in quiet environment; ⩾85% in moderate noise (45-65dB)	Run 100-command test set; calculate WER (Word Error Rate)
Continuous Recognition	Supported (hold mic button for hands-free mode)	Verify auto-stop after 5s silence or command completion
Voice Feedback	Audio confirmation (chime) + Visual LED ring for command execution	Verify audio is not muted when TV is muted (configurable)
Mic Mute	Physical switch or long-press Voice button	Verify hardware mute disables mic power rail; verify LED Red

16. HDMI-CEC (SimpLink) & Universal Control Specification

16.1 HDMI-CEC (SimpLink) Features

Feature	Description	Expected Result	Test Condition
One Touch Play	Insert HDMI device and power on	TV automatically switches to that HDMI input and powers on	Device sends `<Active Source>` message
System Standby	Power off TV with remote	All connected CEC devices also power off (if supported)	TV sends `<Standby>` to all active CEC devices
System Audio Mode	TV audio output to soundbar via HDMI-CEC	TV mute/volume controls the soundbar volume directly	Verify TV OSD shows Soundbar volume, not TV volume
Input Switch	Select HDMI source on external device	TV remote controls external device input source	Verify CEC `<Routing Change>` or `<Set Stream Path>`
Timer Control	Set sleep timer on external device	External device respects TV’s sleep timer	Verify external device powers off when TV sleeps
OSD Transfer	External device menu display	External device menu appears on TV screen	Verify no double OSD (TV + Device)

16.2 Universal Remote (IR Learning)

Feature	Description	Expected Result
Learn IR Signal	Learn IR codes from original remote	Remote stores learned IR codes in internal non-volatile memory
Device Type	TV, DVD, VCR, Cable Box, Sound System, PC	Remote sends correct IR protocol (NEC, RC5, RC6, Sony SIRC) for selected device type
Brand Code	Auto-detect or manual brand code entry	Remote sends correct brand-specific IR codes from database
Code Library	15,000+ device codes in database	Coverage for major brands (Sony, Samsung, LG, Panasonic, Pioneer, etc.)
Learning Range	5m line-of-sight	IR learning works within 5m; verify signal strength > −30dBm
Memory	5 device slots	Can learn up to 5 external devices simultaneously; verify overwrite protection

16.3 SimpLink (HDMI-CEC) Setup

Step	Action	Expected Result
1	Settings > All Settings > General > External Device Manager > SIMPLINK	SIMPLINK settings menu opens
2	Enable SIMPLINK	SIMPLINK enabled; connected devices auto-detected within 5s
3	Device Detection	List of connected CEC devices displayed with Logical Address
4	Device Name Edit	User can rename detected devices; name persists after reboot
5	Power Link	Toggle: Power off TV → Power off all CEC devices; verify delay < 2s

17. Power Management & Battery Specification

17.1 Power Specifications

Parameter	Specification	Test Method
Battery Type	2 × AAA (7 号) Alkaline Battery	Verify compartment fitment; verify polarity protection
Battery Voltage	1.5V × 2 = 3.0V (nominal)	Measure open-circuit voltage with fresh batteries
Low Battery Threshold	2.4V (1.2V per cell)	Use programmable power supply; verify OSD trigger
Critical Battery Threshold	2.1V (1.05V per cell)	Use programmable power supply; verify feature disablement
Battery Life (Typical)	12 months (normal use, ~4 hours/day)	Calculate based on 2500mAh capacity and average current
Battery Life (Pointer Heavy)	6 months (heavy pointer/voice use)	Continuous gyro + BLE streaming test
Battery Life (IR Only)	18 months (pointer/voice disabled)	IR burst test (100 presses/hour)
Power Off Auto-Sleep	Remote enters low-power mode after 30 seconds of inactivity	Verify current drop to < 50μA after 30s idle
Bluetooth Sleep Current	⩽5mA (active connection), ⩽50μA (deep sleep)	Measure with multimeter in series
IR Standby Current	⩽10μA	Measure with multimeter in series

17.2 Low Battery Behavior

Battery Level	Behavior	User Feedback
⩾ 2.8V (Normal)	All functions normal	None
2.4V - 2.8V (Low)	Pointer speed may be reduced; Voice recognition timeout reduced	OSD “Battery Low” notification appears once per session
2.1V - 2.4V (Critical)	Pointer and Voice features disabled; Only IR basic functions work	Red LED blinks on each keypress; OSD “Replace Battery”
< 2.1V (Replace Now)	All functions disabled except Power (IR)	Red LED blinks rapidly (4Hz); OSD “Replace Battery Immediately”

17.3 Battery Replacement Procedure

Step	Action	Expected Result
1	Open battery compartment (slide back cover)	Battery compartment opens; mechanical click heard
2	Remove old batteries	Old batteries removed; verify spring contact integrity
3	Insert 2 × AAA batteries (correct polarity)	Batteries seated correctly; verify polarity diagram matches
4	Close battery compartment	Remote powers on automatically; Blue LED blinks once
5	Press OK wheel	Re-pairing prompt appears on TV if pairing was lost during swap

18. Bluetooth Pairing & Connection Specification

18.1 Pairing Protocol

Parameter	Specification
Protocol	Bluetooth 5.0 (BR/EDR + BLE)
Pairing Method	Press OK wheel to initiate pairing (when unpaired)
Pairing Timeout	10 seconds from pressing OK wheel
Max Paired Devices	4 remotes (TV side); 8 remotes (remote side memory)
Active Connection	1 remote (primary); others in polling mode
Connection Interval	15ms - 30ms (normal), 7.5ms (pointer/ gaming mode)
Transmit Power	⩽4dBm (Bluetooth), ⩽20mW (IR)
Operating Frequency	2.402 GHz - 2.480 GHz (Bluetooth); 38 kHz (IR carrier)
Security	SSP (Secure Simple Pairing) with Numeric Comparison

18.2 Pairing States

State	Description	Transition	LED Indication
Unpaired	Remote not yet paired with any TV	→ Pairing (press OK wheel)	No LED
Pairing	Remote searching for TV (advertising)	→ Paired (TV confirms) / → Unpaired (timeout)	Blue LED 1Hz Blink
Paired	Remote connected to TV (encrypted link)	→ Disconnected (out of range) / → Pairing (deregister)	Blue LED Solid
Disconnected	Bluetooth link lost	→ Reconnecting (auto) / → Pairing (manual)	No LED (IR fallback)
Reconnecting	Auto-reconnect attempt (max 5 retries, 2s interval)	→ Paired (success) / → Unpaired (fail)	Blue LED 2Hz Blink

18.3 Pairing Procedure

Step	Action	Expected Result
1	Insert batteries / Power on remote	Remote powers on; Blue LED blinks if unpaired
2	Point remote at TV (within 3m)	TV detects remote advertising packet
3	Press and release OK wheel	Blue LED starts blinking; TV shows “Magic Remote has been successfully registered”
4	Wait for confirmation (⩽10 seconds)	Pairing complete; LED becomes solid blue
5	Test functions	All buttons and pointer should work; verify pointer calibration

18.4 Re-pairing Procedure

Step	Action	Expected Result
1	Hold Home + Back for 5 seconds	Remote deregisters; LED blinks red 3 times
2	Press OK wheel within 10 seconds	Remote enters pairing mode again; Blue LED blinks
3	Point at TV and wait	TV shows pairing confirmation
4	Pairing complete	All functions restored; LED solid blue

18.5 Connection Troubleshooting

Issue	Possible Cause	Solution
No pairing response	Batteries low	Replace batteries
	Remote out of range	Move within 3m of TV
	TV Bluetooth disabled	Enable Bluetooth in TV settings
	Interference from Wi-Fi/ USB 3.0	Move interfering devices away; use USB 2.0 extension
Intermittent connection	Weak signal	Replace batteries; check for obstructions (metal stands)
	Firmware outdated	Update TV firmware to latest version
Pairing fails	Too many paired remotes	Deregister unused remotes in TV settings
	TV firmware bug	Restart TV (Power cycle); update firmware
Pointer lag	High connection interval	Move closer to TV; reduce Wi-Fi interference

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19. LED Indicator & Haptic Feedback Specification

19.1 LED